

Abhishek Ingle

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PROFESSIONAL EXPERIENCE

Research Software Engineer, Human Robot Interaction Lab | Indiana University Bloomington, IN

Jan 2025 - June 2025

Environment: Python, ROS, Docker, React Native, Next.js, Figma, Vercel, cross-functional

- Drove a 25% lift in user engagement on Honda's *Haru* robot by continually refining real-time social behaviors in Python/ROS and shipping containerized updates with Docker.
- Championed a 30% reduction in cloud infrastructure costs by migrating on-premise ML workloads to a Kubernetes-orchestrated, GPU-accelerated AWS cluster.
- Optimized distributed training of large language models on a multi-GPU cluster, improving model convergence time by 15% through meticulous performance profiling and bottleneck identification.
- Designed and implemented robust CI/CD pipelines for AI services on AWS, ensuring 99.9% uptime and seamless deployments.

Software Engineer (Contract), Nucleo Research Inc. | San Francisco, CA

Jul 2024 - Sep 2024

Environment: OpenAI API, LangChain, Notion API, Next.js, Flask, Anthropic API, Docker, AWS, Google OAuth, FastAPI

- Led a 4-person team to launch an AI productivity assistant syncing Notion, Drive, and OneDrive with 100% data consistency.
- Automated data synchronization and task management workflows using Python scripts, reducing manual task time by 48%.
- Reduced API latency by 48% by redesigning service logic in Next.js/Node and implementing Redis caching, enabling the system to support 2x user growth without performance loss.
- Cut inference time by 40% by designing a memory-aware context router with LangChain and FastAPI, selecting FastAPI for its async support to handle high-concurrency workloads.
- Improved query accuracy 25% and response time 40% by modelling data in PostgreSQL + GraphQL and tuning indexes.

Software Engineer Intern (Remote), Headstarter AI | New York City, NY

Jul 2024 - Sep 2024

Environment: RAG, Llama 3.1, sentiment analysis, scikit-learn, cloud computing, Heroku, REST APIs, CI/CD pipelines, Redis

- Configured cloud VPC networking, firewalls, and port forwarding rules in AWS/GCP deployments for real-time services.
- Generated 500+ AI-curated recipes on AWS EC2/S3, trimming content-ops costs 20% through batch data workflows.
- Utilized Wireshark, netstat, and curl to debug network bottlenecks in microservices-based infrastructure.
- Diagnosed and resolved service outages by analyzing Splunk logs, CloudWatch metrics, and Prometheus alerts.

PROJECTS

Project Keystone | iOS + Android | Typescript, React Native, GCP, Agile Development, Customer Empathy

[Demo Link](#)

- A mobile application designed to combat doomscrolling by locking the user's phone until a specified task is completed. Users submit a photo as proof of completion, which is analyzed by a vision model for approval. A safety lock feature freezes the screen, enforcing focus on the task.

Luna.ai | Web | Next.js 14, OpenAI API, MUI, open source, Database, UX Design

[Live Link](#)

- An accessible platform designed to educate kindergarten and high school students about outer space. It features simplified physics simulations and an AI chatbot for answering cosmic inquiries, utilizing three.js for advanced 3D visualizations.

Japanese Education Platform | Web + Android | React Native, Typescript, Jest/RTL, Dart, Langchain

[Github](#)

- Created a Japanese pronunciation learning app with React Native and Node.js, offering real-time user feedback. Utilized the ElevenLabs API for high-quality text-to-speech and Azure Speech Service for AI-driven pronunciation analysis. Developed the server-side audio processing pipeline using Express and FFmpeg to accurately score pronunciation and manage audio format conversion.

TECHNICAL SKILLS

Languages: Python, JavaScript, C++, C#, Java, Go

Cloud / DevOps: Kubernetes, Docker, AWS Lambda, Jenkins, CI/CD Pipelines, Azure ML, Google Cloud Platforms

Libraries & Frameworks: React.js, Node.js, Express.js, Angular, Flask, Django, Socket.IO, Spark, TensorFlow, PyTorch, Scikit-learn

Databases & Web Technologies: PostgreSQL, NoSQL, Redis, GraphQL, RESTful APIs, ElastiCache, OAuth/OpenID, DynamoDB

System Architecture & Tools: Microservices, Apache Kafka, Git, JIRA, Selenium, linux, software development life cycle, problem solving

EDUCATION

Masters in Computer Science - Indiana University Bloomington

Aug 2023 - May 2025

Data Structures, Operating Systems, Data Analytics, data science, Web Development

Bachelors in Computer Engineering - Pune University, India

Jul 2019 - May 2023

Distributed Systems, Data Analytics Honor, Product Development